

Администрирование локальных сетей

Лабораторная работа №4 — Первоначальное конфигурирование сети

Абдуллахи Бахара

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Российский университет дружбы народов, Москва, Россия

Цель работы

Провести подготовительную работу по первоначальной настройке коммутаторов сети и подготовить устройства к удалённому администрированию по SSH.

Выполнение лабораторной работы

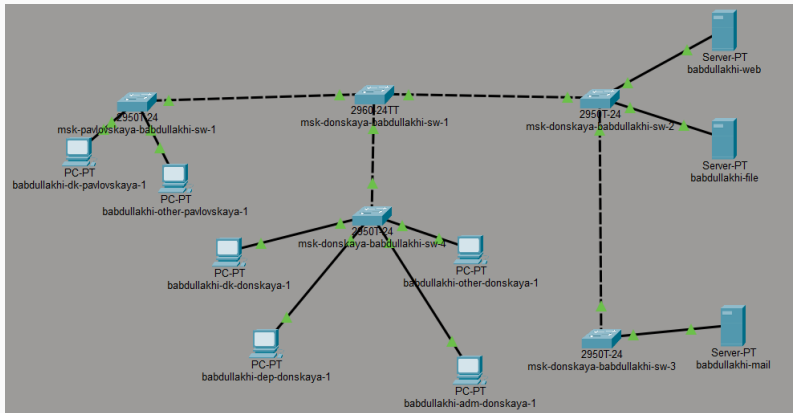


Рис. 1: Топология сети L1

На каждом коммутаторе выполнены действия: - настройка hostname; - создание интерфейса VLAN 2 и включение порта; - назначение IP-адреса управления; - настройка ip default-gateway; - пароли для console и vty; - enable secret и service password-encryption; - пользователь admin; - доменное имя; - генерация RSA-ключей; - разрешение доступа по SSH (transport input ssh); - сохранение конфигурации (write memory).

Настройка коммутаторов

```

Switch>en
Switch#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-donskaya-babdullakhi-sw-1
msk-donskaya-babdullakhi-sw-1(config)#int vlan2
msk-donskaya-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-babdullakhi-sw-1(config-if)#no sh
msk-donskaya-babdullakhi-sw-1(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-babdullakhi-sw-1(config-if)#exit
msk-donskaya-babdullakhi-sw-1(config)#ip defa
msk-donskaya-babdullakhi-sw-1(config)#ip default-gateway 10.128.1.1
msk-donskaya-babdullakhi-sw-1(config)#line vty 0 4 pass cisco
^
% Invalid input detected at '^' marker.

msk-donskaya-babdullakhi-sw-1(config)#line vty 0 4
msk-donskaya-babdullakhi-sw-1(config-line)#pass cisco
msk-donskaya-babdullakhi-sw-1(config-line)#login
msk-donskaya-babdullakhi-sw-1(config-line)#exit
msk-donskaya-babdullakhi-sw-1(config)#line console 0
msk-donskaya-babdullakhi-sw-1(config-line)#pass cisco
msk-donskaya-babdullakhi-sw-1(config-line)#login
msk-donskaya-babdullakhi-sw-1(config-line)#exit
msk-donskaya-babdullakhi-sw-1(config)#enab sec cisco
msk-donskaya-babdullakhi-sw-1(config)#serv pass
msk-donskaya-babdullakhi-sw-1(config)#user admin priv 1 sec cisco
msk-donskaya-babdullakhi-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-babdullakhi-sw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-babdullakhi-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-babdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:9:28.582: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:9:28.582: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-babdullakhi-sw-1(config-line)#trans in ssh
msk-donskaya-babdullakhi-sw-1(config-line)#exit

```

```

Switch>enab
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-donskaya-babdullakhi-sw-2
msk-donskaya-babdullakhi-sw-2(config)#int vlan2
msk-donskaya-babdullakhi-sw-2(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-babdullakhi-sw-2(config-if)#no sh
msk-donskaya-babdullakhi-sw-2(config-if)#ip addr 10.128.1.3 255.255.255.0
msk-donskaya-babdullakhi-sw-2(config-if)#exit
msk-donskaya-babdullakhi-sw-2(config)#ip def 10.128.1.1
msk-donskaya-babdullakhi-sw-2(config)#line vty 0 4
msk-donskaya-babdullakhi-sw-2(config-line)#pass cisco
msk-donskaya-babdullakhi-sw-2(config-line)#login
msk-donskaya-babdullakhi-sw-2(config-line)#exit
msk-donskaya-babdullakhi-sw-2(config)#line console 0
msk-donskaya-babdullakhi-sw-2(config-line)#pass cisco
msk-donskaya-babdullakhi-sw-2(config-line)#login
msk-donskaya-babdullakhi-sw-2(config-line)#exit
msk-donskaya-babdullakhi-sw-2(config)#enab sec cisco
msk-donskaya-babdullakhi-sw-2(config)#serv pass
msk-donskaya-babdullakhi-sw-2(config)#user admin priv 1 sec cisco
msk-donskaya-babdullakhi-sw-2(config)#ip domain-nam
msk-donskaya-babdullakhi-sw-2(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-babdullakhi-sw-2(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-babdullakhi-sw-2.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-babdullakhi-sw-2(config)#line vty 0 4
*Mar 1 0:13:1.458: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:13:1.458: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-babdullakhi-sw-2(config-line)#trans in ssh
msk-donskaya-babdullakhi-sw-2(config-line)#exit
msk-donskaya-babdullakhi-sw-2(config)#exit
msk-donskaya-babdullakhi-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

```

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-donskaya-babdullakhi-sw-3
msk-donskaya-babdullakhi-sw-3(config)#int vlan2
msk-donskaya-babdullakhi-sw-3(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-babdullakhi-sw-3(config-if)#no sh
msk-donskaya-babdullakhi-sw-3(config-if)#ip addr 10.128.1.4 255.255.255.0
msk-donskaya-babdullakhi-sw-3(config-if)#exit
msk-donskaya-babdullakhi-sw-3(config)#ip def 10.128.1.1
msk-donskaya-babdullakhi-sw-3(config)#line vty 0 4
msk-donskaya-babdullakhi-sw-3(config-line)#pass cisco
msk-donskaya-babdullakhi-sw-3(config-line)#login
msk-donskaya-babdullakhi-sw-3(config-line)#exit
msk-donskaya-babdullakhi-sw-3(config)#line cons 0
msk-donskaya-babdullakhi-sw-3(config-line)#pass cisco
msk-donskaya-babdullakhi-sw-3(config-line)#login
msk-donskaya-babdullakhi-sw-3(config-line)#exit
msk-donskaya-babdullakhi-sw-3(config)#enab sec cisco
msk-donskaya-babdullakhi-sw-3(config)#serv pas
msk-donskaya-babdullakhi-sw-3(config)#user admin priv 1 sec cisco
msk-donskaya-babdullakhi-sw-3(config)#ip dom
msk-donskaya-babdullakhi-sw-3(config)#ip doma
msk-donskaya-babdullakhi-sw-3(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-babdullakhi-sw-3(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-babdullakhi-sw-3.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
    a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-babdullakhi-sw-3(config)#line vty 0 4
*Mar 1 0:15:44.875: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:15:44.875: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-babdullakhi-sw-3(config-line)#trans in ssh
msk-donskaya-babdullakhi-sw-3(config-line)#exit
msk-donskaya-babdullakhi-sw-3(config)#exit
msk-donskaya-babdullakhi-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

```

```

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-donskaya-babdullakhi-sw-4
msk-donskaya-babdullakhi-sw-4(config)#int vlan2
msk-donskaya-babdullakhi-sw-4(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-babdullakhi-sw-4(config-if)#no sh
msk-donskaya-babdullakhi-sw-4(config-if)#ip addr 10.128.1.5 255.255.255.0
msk-donskaya-babdullakhi-sw-4(config-if)#exit
msk-donskaya-babdullakhi-sw-4(config)#ip def 10.128.1.1
msk-donskaya-babdullakhi-sw-4(config)#line vty 0 4
msk-donskaya-babdullakhi-sw-4(config-line)#pass cisco
msk-donskaya-babdullakhi-sw-4(config-line)#login
msk-donskaya-babdullakhi-sw-4(config-line)#exit
msk-donskaya-babdullakhi-sw-4(config)#line cons 0
msk-donskaya-babdullakhi-sw-4(config-line)#pass cisco
msk-donskaya-babdullakhi-sw-4(config-line)#login
msk-donskaya-babdullakhi-sw-4(config-line)#exit
msk-donskaya-babdullakhi-sw-4(config)#enab sec cisco
msk-donskaya-babdullakhi-sw-4(config)#serv pass
msk-donskaya-babdullakhi-sw-4(config)#user admin priv 1 sec cisco
msk-donskaya-babdullakhi-sw-4(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-babdullakhi-sw-4(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-babdullakhi-sw-4.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-babdullakhi-sw-4(config)#line vty 0 4
*Mar 1 0:17:49.74: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:17:49.74: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-babdullakhi-sw-4(config-line)#trans in ssh
msk-donskaya-babdullakhi-sw-4(config-line)#exit
msk-donskaya-babdullakhi-sw-4(config)#exit
msk-donskaya-babdullakhi-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-sw-4#wr mem
Building configuration...
[OK]
msk-donskaya-babdullakhi-sw-4#

```

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-pavlovskaya-babdullakhi-sw-1
msk-pavlovskaya-babdullakhi-sw-1(config)#int vlan2
msk-pavlovskaya-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-pavlovskaya-babdullakhi-sw-1(config-if)#no sh
msk-pavlovskaya-babdullakhi-sw-1(config-if)#ip addr 10.128.1.6 255.255.255.0
msk-pavlovskaya-babdullakhi-sw-1(config-if)#exit
msk-pavlovskaya-babdullakhi-sw-1(config)#ip def 10.128.1.1
msk-pavlovskaya-babdullakhi-sw-1(config)#line vty 0 4
msk-pavlovskaya-babdullakhi-sw-1(config-line)#pass cisco
msk-pavlovskaya-babdullakhi-sw-1(config-line)#login
msk-pavlovskaya-babdullakhi-sw-1(config-line)#exit
msk-pavlovskaya-babdullakhi-sw-1(config)#line cons 0
msk-pavlovskaya-babdullakhi-sw-1(config-line)#pass cisco
msk-pavlovskaya-babdullakhi-sw-1(config-line)#login
msk-pavlovskaya-babdullakhi-sw-1(config-line)#exit
msk-pavlovskaya-babdullakhi-sw-1(config)#enab sec cisco
msk-pavlovskaya-babdullakhi-sw-1(config)#serv pass
msk-pavlovskaya-babdullakhi-sw-1(config)#user admin priv 1 sec cisco
msk-pavlovskaya-babdullakhi-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-pavlovskaya-babdullakhi-sw-1(config)#crypto key gen rsa
The name for the keys will be: msk-pavlovskaya-babdullakhi-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-babdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:20:47.234: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:20:47.234: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-pavlovskaya-babdullakhi-sw-1(config-line)#trans in ssh
msk-pavlovskaya-babdullakhi-sw-1(config-line)#exit
msk-pavlovskaya-babdullakhi-sw-1(config)#exit
msk-pavlovskaya-babdullakhi-sw-1#
%SYS-5-CONFIG I: Configured from console by console

```

Результаты

В ходе лабораторной работы: - построена топология сети уровня L1 в Cisco Packet Tracer; - выполнена первоначальная настройка 5 коммутаторов по единому шаблону; - назначены IP-адреса управления на интерфейсе VLAN 2; - настроены пароли и включён защищённый доступ по SSH; - конфигурации сохранены в память устройств.

Коммутаторы готовы к удалённому администрированию в сети управления 10.128.1.0/24.